

Lab Manual: Managing Secrets and Packaging Streamlit Apps

Objective

By the end of this lab, you will:

- Securely store API keys and credentials using `.streamlit/secrets.toml`.
- Configure app themes using `.streamlit/config.toml`.
- Prepare your app for **sharing and deployment** with `requirements.txt`.

Prerequisites

- Python 3.7+ installed.
- Streamlit installed: `pip install streamlit requests`
- A basic Streamlit app (`app.py`).

Step 1: Securely Store Secrets

1. Inside your project folder, create a hidden directory:

```
.streamlit/
```

2. Inside it, create a file named `secrets.toml`:

```
API_KEY = "your_api_key_here"  
USERNAME = "your_username"  
PASSWORD = "your_password"
```

3. **Important:** Add `.streamlit/secrets.toml` to your `.gitignore` file so it is not pushed to GitHub or shared publicly:

```
.streamlit/secrets.toml
```

This keeps your secrets safe from version control.

Step 2: Access Secrets in Your App

Update `app.py`:

```
import streamlit as st
import requests

# Access secret values
api_key = st.secrets["API_KEY"]

st.title("Secure API Access Demo")

# Example API request
url = "https://api.example.com/data"
headers = {"Authorization": f"Bearer {api_key}"}

response = requests.get(url, headers=headers)

if response.status_code == 200:
    st.write("API Response:", response.json())
else:
    st.error(f"Error: {response.status_code}")
```

Explanation:

- `st.secrets["API_KEY"]` → securely loads the value from `secrets.toml`.
- No keys are hardcoded in the source code.
- If someone else runs the app, they must create their own `secrets.toml`.

Step 3: Configure App Theme

Create a file `.streamlit/config.toml`:

```
[theme]
primaryColor = "#FF4B4B"
backgroundColor = "#FFFFFF"
secondaryBackgroundColor = "#F0F2F6"
textColor = "#262730"
font = "sans serif"
```

Explanation:

- `primaryColor` → main highlight color (buttons, sliders).
- `backgroundColor` → page background.
- `textColor` → default font color.
- `font` → font style for the app.

Ensures your app looks consistent locally and in the cloud.

Step 4: Prepare Dependencies

Create a file `requirements.txt`:

```
streamlit
requests
```

This lists all dependencies needed to run the app.

When collaborators or cloud services install the app, they just run: `pip install -r requirements.txt`

Step 5: Run the App Locally

Start your app: `streamlit run app.py`

Your app will load at <http://localhost:8501>.

Step 6: Share or Deploy

- **Collaborators:** They can clone your repo, install dependencies from `requirements.txt`, and create their own `.streamlit/secrets.toml` with their keys.
- **Deployment:** On platforms like **Streamlit Cloud**, you upload your repo, and add secrets in the web UI instead of uploading `secrets.toml`.

Verification Checklist

- `secrets.toml` created and ignored in `.gitignore`.
- App successfully accessed secrets with `st.secrets`.
- Configured theme using `config.toml`.
- `requirements.txt` lists all dependencies.
- App runs locally with `streamlit run app.py`.
- Code is ready for collaboration or deployment.

Summary

In this lab, you:

- Secured sensitive values with `.streamlit/secrets.toml`.
- Configured your app's look using `config.toml`.
- Prepared your app for **sharing or deployment** with `requirements.txt`.

With these steps, your Streamlit apps are now **secure, professional, and ready for the cloud.**